The polar-pentagon perimeter centroid has degree at least four

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Abstract

We give an exact finite certificate excluding every cubic from the perimeter-centroid locus of the polar pentagons in the focus-circle Poncelet family. Algebraicity follows from the tangent recurrence and the edge-length square roots. Together with the previous general polar-centroid theorem, this resolves the intended pentagon conjecture of Bellio, Garcia and Reznik.

Task 1: the original problem and its intended scope

Consider the convex Poncelet pentagons inscribed in a parabola of focal distance $f > 0$ and circumscribed about a circle centered at its focus. Their polar polygons have as sides the tangents to the parabola at the five original vertices. The perimeter centroid is the centroid of their five segments with positive Euclidean length as mass, including overlapping segments with multiplicity.

Conjecture 3 of [1] asserts straight vertex- and area-centroid loci and perimeter-centroid degree at least four. Its word *perpendicular* before *directrix* is a direction error: the intended direction is parallel, as stated in its general Conjecture 5. We prove the substantive degree assertion, rather than count that wording error as a counterexample.

Theorem 1. *The perimeter-centroid locus is an algebraic locus of degree at least four: it is contained in an algebraic plane curve, but no nonzero polynomial of total degree at most three vanishes on it. The vertex- and area-centroid loci are complete lines parallel to the directrix.*

Here degree means the degree of the algebraic closure of the physical locus. The positive length convention selects real branches; as in the source's quadrilateral example, an implicit equation can have additional real points. We claim the lower bound, not an exact degree or an equation with no extraneous branches.

The line assertions are the $N = 5$ case of the previously established general theorem [2]; they receive no additional credit. In coordinates $y^2 + 4fx = 0$, they are $x = h_0$ and $x = (8f^2 - r^2)/(12f)$, respectively. The new proof below establishes the independent perimeter-degree assertion.

Algebraic coordinates and the six samples

A similarity permits $f = 1$. By [1, Proposition 25], the required radius is the unique positive root r of

$$r^6 + 12r^5 - 28r^4 + 32r^3 + 112r^2 - 64r - 64 = 0, \qquad \frac{995}{1000} < r < \frac{996}{1000}. \tag{1}$$

It is approximately 0.995219078669842. Write $P(t) = (-t^2/4, t)$. The chord through $P(t), P(s)$ is tangent to the focus circle if and only if

$$(ts + 4)^2 = r^2\big(16 + (t + s)^2\big). \tag{2}$$

Indeed, the chord is $4x + (t+s)y - ts = 0$ and the focus is $(-1, 0)$.

For a starting value a, put $t_0 = a$ and define

$$A = a^2 - r^2, \qquad B = (8 - 2r^2)a, \qquad C = 16 - 16r^2 - r^2a^2,$$

$$t_1 = \frac{-B + \sqrt{B^2 - 4AC}}{2A}, \tag{3}$$

$$t_{j+1} = -\frac{(8 - 2r^2)t_j}{t_j^2 - r^2} - t_{j-1}, \qquad j = 1, 2, 3. \tag{4}$$

The positive square root in (3) chooses one traversal direction. Equation (4) is Vieta's formula for the other root of (2). The cited closure theorem makes the resulting five vertices a closed Poncelet pentagon.

The intersection of the tangents at parameters t_i, t_{i+1} is

$$Q_i = \left(-\frac{t_i t_{i+1}}{4}, \frac{t_i + t_{i+1}}{2} \right), \qquad i \pmod 5. \tag{5}$$

Consequently, with positive square roots throughout,

$$\ell_i = \sqrt{(Q_{i+1,x} - Q_{i,x})^2 + (Q_{i+1,y} - Q_{i,y})^2}, \qquad (X, Y) = \frac{\sum_{i=0}^{4} \ell_i (Q_i + Q_{i+1})}{2 \sum_{i=0}^{4} \ell_i}. \tag{6}$$

These formulas show algebraicity. The two possible values of t_1 form a finite algebraic extension of $\mathbb{R}(a)$; the other parameters and polar vertices are rational in it. Adjoining the five square roots in (6) is again a finite algebraic extension. Its transcendence degree is one, so X, Y satisfy a nonzero polynomial relation. There are only finitely many branches, so multiplying their relations gives one relation for the entire locus. Exceptional parameter values are covered by continuity whenever the polygon and its centroid are finite.

Use the following six *exact rational* starting values:

$$a_1 = \frac{3}{25}, \quad a_2 = \frac{27}{100}, \quad a_3 = \frac{43}{100}, \quad a_4 = \frac{17}{25}, \quad a_5 = \frac{131}{100}, \quad a_6 = \frac{12}{5}. \tag{7}$$

Let (X_i, Y_i) be defined exactly by (1)–(6). For orientation only, their approximate coordinates are

i	X_i	Y_i
1	0.531703184427	9.271732856495
2	0.501537653846	−1.067231332343
3	0.496287921973	0.232794002460
4	0.509767562441	2.074695123461
5	0.517049614999	−3.248106062772
6	0.496610704943	−0.328686222928

The rounded entries in this table are not used in the certificate.

An exact determinant certificate

Form the matrix with rows

$$M_i = (1,\ X_i,\ X_i^2,\ X_i^3,\ Y_i^2,\ X_i Y_i^2).$$

Lemma 1. *All six sample polygons are finite, strictly convex members of the required family; all Y_i are nonzero; and*

$$-\frac{1769}{10^{18}} < \det M < -\frac{1768}{10^{18}}. \tag{8}$$

Proof. The accompanying `verify.py` is an exact certificate evaluator using only integers and rational numbers from the Python standard library. The algorithm and its bounds are specified here so that the computation does not depend on floating-point behavior.

Set $S = 2^{512}$ and represent $[L/S, U/S]$ by the integers (L, U). For addition and negation use the exact endpoint formulas. For multiplication or division, take the minimum and maximum of all four endpoint products or quotients and round outwards to multiples of $1/S$. Every division asserts that its denominator interval excludes zero. For a nonnegative interval, the square-root endpoints in integer units are

$$\left\lfloor \sqrt{LS} \right\rfloor, \qquad \left\lceil \sqrt{US} \right\rceil.$$

Integer square root and an exact square comparison implement them. Squaring uses zero as the lower endpoint when the input crosses zero. These rules preserve enclosure at every operation.

The polynomial in (1) has opposite signs at 995/1000 and 996/1000. Evaluating its derivative by interval arithmetic on this whole interval gives a strictly positive lower bound, so the root there is unique. Exact rational bisection refines the root bracket to width below 2^{-520}; its outward dyadic enclosure is used in every sample.

Evaluate (3)–(6), asserting nonzero denominators, positive radicands, distinct vertex parameters and $Y_i \neq 0$. Exact signs of edge cross products show that every other original vertex and the focus lie strictly on the same side of each directed edge. Thus the original pentagons are strictly convex with the circle on their interior side. Closure uses Proposition 25, not a small numerical residual. All polar segment lengths are certified positive.

Finally, evaluate the determinant as the signed sum of all $6! = 720$ permutation products with these same interval rules. The resulting enclosure, rounded outwards here to 35 decimal places, is

$$\big[-0.00000000000000176874062193541107867,\ -0.00000000000000176874062193541107866\big].$$

It lies strictly within (8). The full enclosure, sample bounds and all assertions are recorded in `verification-results.json`. This proves the lemma by a finite exact calculation. □

Why the certificate excludes every cubic

Reflection in the parabola's axis preserves the Poncelet family, polarity, and positive Euclidean edge lengths. Hence the locus contains both (X_i, Y_i) and $(X_i, -Y_i)$.

Suppose a polynomial $H(X, Y)$ of total degree at most three vanishes on the entire locus. Its even part in Y has the form

$$H_{\text{even}} = c_0 + c_1 X + c_2 X^2 + c_3 X^3 + c_4 Y^2 + c_5 X Y^2.$$

Adding its values at the reflected samples gives $Mc = 0$. Lemma 1 makes M invertible, so all six coefficients vanish. The remaining odd part is

$$H_{\text{odd}} = Y(d_0 + d_1 X + d_2 X^2 + d_3 Y^2).$$

Since each $Y_i \neq 0$, the six sample equations say that a linear combination of columns $1, 2, 3, 5$ of M is zero. Those columns are independent, again because M is invertible. Thus all four d_j vanish, and $H = 0$.

The algebraic locus therefore has degree at least four. This also proves it is nonconstant. Restoring f applies an invertible similarity, which preserves degree and completes the perimeter assertion for every scale.

Four proposed follow-up tasks

1. **Task 2.** Determine the exact degree and a minimal implicit equation of this perimeter-centroid locus.

2. **Task 3.** Determine its real branches and which parts are attained by finite polar pentagons.

3. **Task 4.** Find exact bounds for its horizontal coordinate and identify all extremizing configurations.

4. **Task 5.** Determine the degree of the polar perimeter-centroid locus for each convex period $N \geq 6$.

References

[1] F. Bellio, R. Garcia, D. Reznik, *Parabola-Inscribed Poncelet Polygons Derived from the Bicentric Family*, KoG **26** (2022), 3–15, doi:10.31896/k.26.1; arXiv:2111.00979v3.

[2] Principia Math In Progress, *Straight centroid loci for polar Poncelet polygons*, MathDB 352456, 21 September 2026. Repository directory `problems/352456`, with the full proof in `solution.tex` and `solution.pdf`, commit `ece807501ca0`.